# Mohammed Abduljabbar

PhD Informatics and Computing

CONTACT

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Google Scholar

## TECHNICAL SKILLS

- Deep Learning
- Computer Vision
- Spatiotemporal Modeling
- Transformer Models
- Robotics & Human Aug.
- ML Pipelines / PyTorch

## SKILLS & EXPERTISE

Python · C++ · Go · PHP · JavaScript · Full Stack · Linux · Git · Kubernetes · DevOps · Cloud Computing (Azure, AWS, GCP, DigitalOcean) · Edge Computing · IoT · Apache · Nginx · NVIDIA DGX-1 · NVIDIA Jetson · Raspberry Pi · Arduino · 3D Modeling and Animation · Unity Engine · Unreal Engine · Blender · GIS · LLM · VLM · vLLM · Ollama · LangChain · LangGraph · Vector Databases · Hugging Face · LoRA · QLoRA · Unsloth · TensorFlow.js · PyTorch · MediaPipe · Cursor · Claude Code · Gemini · ChatGPT · DeepSeek · Qwen · Blockchain · Automated Crypto Spot / Futures Bots · RAG · Agentic AI · Vibe Coding

## LANGUAGES

| | |
|---|---|
| English | ●●●●● |
| Arabic | ●●●●● |
| Japanese | ●○○○○ |
| German | ●○○○○ |

## About Me

PhD in Informatics and Computing with deep expertise in deep learning, spatiotemporal modeling, robotics, and deployable AI systems. Research focuses on lightweight architectures for real-world applications — computer vision, Arabic Sign Language recognition using Transformer-based and state-space (Mamba) models, 3D digital twins for human–machine interaction, and Mediapipe-based real-time perception pipelines.

Professional background spans scalable web platforms, cloud-native systems, AI-powered applications, DevOps, robotics systems, and automated trading infrastructure. Passionate about bridging academic research and industry deployment.

## What I Do

### Deep Learning Research
Designing and training spatiotemporal models, transformers, and state-space architectures for real-world AI.

### Computer Vision
Sign language recognition, 3D digital twins, and real-time Mediapipe perception pipelines.

### Robotics & Augmentation
ROS-based autonomous mobile robotics (Husky platform), humanoid social robotics (NAO, Pepper), brain–computer interface (BCI) systems, cognitive assistive systems for elderly care, and haptic interfaces for space exploration simulations.

### Full Stack Development
Cloud-native platforms, AI-powered web apps, DevOps pipelines, and automated trading systems.

## Experience

Jan 2020 — May 2026
**PhD Fellow, Researcher & Full Stack Developer**
AI, Computer Vision & Robotics Lab — UAEU
Robot explorer for deep wells · Cognitive robotic training for elderly care · Arabic Sign Language Translation · Fine-tuning video transformers for sign language recognition.

Apr 2022 — Dec 2023
**PhD Fellow, Researcher & Full Stack Developer**
Interactive Robots & Digital Media Lab — UAEU
AI & robotics for human augmentation and Mars exploration · Astro Haptic Footwear research grant · AI-driven applications and research platforms.

Jan 2024 — Jun 2024
**Adjunct Instructor**
United Arab Emirates University
Taught Programming Lab I (CSBP121): structured programming, algorithms, arrays, selection & repetition structures.

Sep 2023 — Jan 2024
**Higher Education Teaching Instructor**
United Arab Emirates University
Completed: Center of Excellence in Teaching and Learning (CETL) PhD Teaching Academy (Cohort 8).

Apr 2022 — Jan 2024
**Research Assistant**
Zayed University
Astro Haptic Footwear research grant · AI and robotics research development.

Sep–Nov 2023
**Research Paper Reviewer**
UAEU — IIT'23 Conference
Reviewer for the International Conference on Innovations in Information Technology.

Jan 2018 — Jan 2019
**IT Supervisor & Web Developer**
Mansion Computer Gaming Center
Supervised IT ops across two branches · Developed company platform and POS system.

Dec 2014 — Mar 2015
**Web Developer / Project Coordinator**
AsfarTrip — Asfar Group, Dubai, UAE
Web Developer and Software Engineer at Asfar Group · 4 mos.

Dec 2010 — May 2014
**Research Developer**
KDD Lab — Kansas State University
Web apps and API integration · Python data processing · GIS-related research publications.

Aug 2013 — Dec 2013
**Teaching Assistant / Lab Assistant**
Kansas State University
TA for Introductory Computer Science course · Scratch programming · Assisted students in labs, exams, and assignments · Graded projects, assignments, and exams.

Jul–Nov 2012
**IT Help Desk & BI Analyst**
Zain KSA
Reports via Oracle, Teradata & SAP · BMC Remedy ticketing · TDWI Business Intelligence training.

## Teaching

**Programming Lab I (CSBP121)**
United Arab Emirates University · Adjunct Instructor
Jan 2024 — Jun 2024

**Teaching Assistant / Lab Assistant**
Kansas State University
Aug 2013 — Dec 2013

## Research Grants & Funded Projects

PhD Fellow / Researcher on three interdisciplinary funded projects with a combined project value of **AED 1.43 million**, spanning artificial intelligence, computer vision, robotics, human augmentation, and intelligent human–machine systems.

### Two-Handed Static and Dynamic Arabic Sign Language Translation System Integration Using Spatiotemporal and Machine Translation Models
UAEU Program for Advanced Research (UPAR) · Apr 2022 — Mar 2024

**Role:** PhD Fellow / Researcher · **PI:** Dr. Munkhjargal Gochoo · **Award:** AED 399,300

Contributed to AI-based Arabic Sign Language translation research using spatiotemporal deep learning, computer vision, temporal modelling, and machine translation.

### Astro Haptic Footwear: Human Augmentation for Mars Exploration
UAEU–Zayed University Joint Research Grant · Jan 2022 — Jan 2024

**Role:** PhD Fellow / Researcher · **PI:** Dr. Jose Berengueres · **Award:** AED 360,500

Contributed to interdisciplinary R&D in human augmentation, haptic interfaces, machine learning, sensing, robotics, prototype development, experimentation, and research dissemination.

### Computer-/Robotic-Based Cognitive Training System for Elderly People in the UAE
UAEU Program for Advanced Research (UPAR) · Jan 2019 — Dec 2022

**Role:** PhD Fellow / Researcher · **PI:** Dr. Fady Shibata Alnajjar · **Award:** AED 673,000

Contributed to AI-enabled cognitive training and autonomous assessment using humanoid social robotics, machine learning, human–robot interaction, and culturally adapted Arabic interactive systems.

## Education

**PhD in Informatics and Computing**
United Arab Emirates University
Jan 2021 — May 2026 · GPA 3.45/4.00

**MSc in Software Engineering**
United Arab Emirates University
Aug 2019 — Dec 2020 · GPA 3.39/4.00

**Postgraduate Courses in Cloud Computing**
Dublin City University
Jan 2015 — Jan 2017

**BSc in Computer Information Systems — Minor: Business Admin**
Kansas State University
Jun 2008 — Dec 2011 · GPA 3.15/4.00

## Achievements

- ★ IEEE SMS BR41N Competition — Brain Computer Interface Hackathon **1st Place**
- ★ ACM Programming Competition **2nd Place**
- ★ Arabic Sign Language AI System Demo — **Abu Dhabi 6G Summit**
- ★ Robot Programmer & Operator — **UAEU Pavilion, EXPO Dubai**
- ★ Astrohaptic Project & Space Suit Demonstrations — **NAJAH Education Expo**

## References

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